Predicting 4th Down Conversion Probability in the NFL

Logan Calder (lcalder@berkeley.edu), Jordan Rivera (jordanrivera@berkeley.edu),
Emily Silvershein (esilvershein@berkeley.edu), Ariel Gholar (agholar@berkeley.edu)

Motivation:

As all four of us have a deep interest in sports, we found the concept of studying NFL data interesting. Particularly, we're interested in what offenses do on 4th down. When facing a 4th down, teams must choose whether to go for it, punt, or attempt a field goal. The risk of going for it is high, as failing to convert results in a turnover. Therefore, a key input into this decision is the probability that the offense will successfully convert if they choose to go for it.

The problem is not simply "Can a team convert?" but rather, "Are teams making the right decisions to maximize their chances of winning?" Coaches often struggle with incredibly difficult choices on 4th down, as the choice could either win or lose a game. We want to evaluate the quality of the decision rather than just its outcome.

We aim to develop an ML framework that estimates 4th down conversion probability based on pre-snap information. In addition to evaluating how accurately we can estimate conversion probability, we are interested in how these predictions could inform a break-even decision framework for determining when going for it is optimal for maximizing win probability.

Data:

We'll use NFL play-by-play data from nflverse, accessed in Python through the nflreadpy package. This package provides a huge amount of NFL data, including play-by-play data dating back to 1999. We plan to restrict the dataset to fourth-down plays in which the offense attempted to convert. Each observation will represent one fourth-down attempt, with the outcome indicating whether the offense successfully converted the attempt.

There are over 370 columns in the play-by-play dataset, and more available in other nflverse datasets. We expect to restrict our predictor variables to information available pre-play, such as yards-to-go, field position, quarter, time remaining, score, and information about the offensive and defensive teams/players. We may also incorporate formation or personnel information.

Related Work:

Previous work has demonstrated how predictive modeling can support fourth-down decision-making. The NFL's Next Gen Stats Decision Guide uses machine-learning models to estimate both fourth-down conversion probability and win probability, combining these estimates to evaluate whether a team should go for it, punt, or attempt a field goal. Similarly, Blinkoff et al. developed a logistic regression model using college football data to predict fourth-down

conversion probability based on factors including yards to go, field position, and play type. Our project builds on these approaches using NFL play-by-play data to estimate conversion probability from pre-snap information and connect those predictions to win probability. This will allow us to evaluate not only whether teams successfully convert, but also whether their fourth-down decisions align with model-estimated strategies for maximizing win probability.

https://www.nfl.com/news/introducing-the-next-gen-stats-decision-guide-a-new-analytics-tool-for-fourth-do

https://thesportjournal.org/article/predictive-modeling-of-4th-down-conversion-in-power-5-conferences-football-data-analytics

https://arxiv.org/abs/2311.03490

GitHub Link: https://github.com/logancalder/DATASCI207-Project